

ATOMIC STRUCTURE

Meaning of Atoms and Molecules

Atom: The smallest fundamental unit of a chemical element that retains the chemical properties of that element. Think of atoms as the basic "bricks" that make up everything around us.

Molecule: A group of two or more atoms chemically bonded together. Molecules can be made of the same type of atom (e.g., O_2 - an oxygen molecule) or different types of atoms (e.g., H_2O - a water molecule). They are the smallest unit of a compound that can exist independently and retain its chemical properties.

Sizes of Atoms and Molecules

Atoms and molecules are incredibly small, far too small to be seen with the naked eye or even most common microscopes.

Atomic sizes(radius) are typically measured in units like nanometers ($1\text{ nm} = 10^{-9}\text{ meters}$) or picometers ($1\text{ pm} = 10^{-12}\text{ meters}$).

For perspective: If a typical atom were scaled up to the size of a sports stadium, a single molecule would be considerably smaller, like a car or a house within that stadium.

Electrons, Protons, and Neutrons

These are the three main subatomic particles that make up an atom:

Particle	Charge	Location	Mass (Approx.)
Proton	Positive (+1)	Nucleus (Center)	1 amu
Neutron	Neutral (0)	Nucleus (Center)	1 amu
Electron	Negative (-1)	Electron Cloud (Orbit)	Very small (negligible for mass)

The **number of protons** in an atom's nucleus determines which element it is.

In a neutral atom, the number of electrons equals the number of protons.

Concept of Atomic Mass

Atomic Mass: The mass of an atom.

It is primarily determined by the total number of **protons and neutrons** in the atom's nucleus (the nucleons).

The mass of electrons is so tiny that it's usually ignored when calculating atomic mass.

Atomic mass is typically measured in Atomic Mass Units (amu).

Concept of Radioactive Elements

Radioactive Elements: Elements whose atoms have unstable nuclei.

Unstable nuclei spontaneously release energy and/or particles in an attempt to become more stable. This emission is called radiation.

Examples of radioactive elements include Uranium (U), Radium (Ra), and Plutonium (Pu).

Radioactive Elements Decay Processes

Radioactive Decay: The natural, spontaneous process by which an unstable atomic nucleus loses energy by emitting radiation.

During decay, the original radioactive atom (parent nuclide) transforms into a different atom (daughter nuclide), which may be stable or still radioactive.

Common types of radiation emitted are alpha particles (α), beta particles (β), and gamma rays (γ).

1. Alpha Decay (α)

This is the "heavy hitter" of decay. The nucleus ejects an alpha particle, which is essentially a Helium nucleus.

- **What is lost:** 2 protons and 2 neutrons.
- **Result:** The atom transforms into a completely different element with an atomic number decreased by 2 and a mass decreased by 4.
- **Power:** These particles are large and slow. They can be stopped by a simple **sheet of paper** or even the outer layer of your skin.

2. Beta Decay (β)

This is a bit of a "magic trick" inside the nucleus. In the most common form (β^-), a neutron turns into a proton and spits out a high-energy electron (the beta particle).

- **What is lost:** A high-speed electron (and an antineutrino).
- **Result:** The atomic number *increases* by 1 (because you gained a proton), but the mass stays the same.
- **Power:** These are smaller and faster than alpha particles. You'll need a **sheet of aluminum** or a thin pane of glass to stop them.

3. Gamma Decay (γ)

Unlike the other two, gamma decay doesn't involve spitting out chunks of matter. Instead, it's a release of pure, high-frequency electromagnetic energy.

- **What is lost:** Energy (photons), but no protons or neutrons.
- **Result:** The element stays the same, it just moves from an "excited" high-energy state to a lower, more stable state.
- **Power:** This is the most "ghost-like" radiation. It can pass through the human body easily and requires **thick lead or several feet of concrete** to be blocked.